



JEPPIAAR INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

“Self-Belief | Self-Discipline | Self-Respect”

Kunnam, Sunguvarchatram, Sriperumbudur – 631 604.



MINOR DEGREE

AGEM302 – SEARCH ENGINE OPTIMIZATION

LAB MANUAL

NAME :

REG NO :

YEAR : III

SEMESTER : V



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VISION AND MISSION OF THE INSTITUTION

INSTITUTE VISION

- ❖ Jeppiaar Institute of Technology aspires to provide technical education in futuristic technologies with the perspective of innovative, industrial, and social applications for the betterment of humanity.

INSTITUTE MISSION

- ❖ To produce competent and disciplined high-quality professionals with the practical skills necessary to excel as innovative professionals and entrepreneurs for the benefit of society.
- ❖ To improve the quality of education through excellence in teaching and learning, research, leadership, and by promoting the principles of scientific analysis, and creative thinking.
- ❖ To provide excellent infrastructure, serene, and stimulating environment that is most conducive to learning.
- ❖ To strive for productive partnership between the Industry and the Institute for research and development in the emerging fields and creating opportunities for employability.
- ❖ To serve the global community by instilling ethics, values, and life skills among the students needed to enrich their lives.

DEPARTMENT VISION

- ❖ To provide inclusive growth by imparting professional education to achieve a center of excellence by developing innovative, industrial and social entrepreneurs, proactive leaders and Critical thinkers and managers through continuous learning towards business management and Research for the betterment of the humanity.

DEPARTMENT MISSION

M1:To develop competent, disciplined and high quality professionals by providing a value- based education and imparting necessary skills to excel in their profession and in entrepreneurship for the benefit of the society

M2:To improve managerial, leadership and entrepreneurial skills by involving in teaching- learning process and in research through scientific analysis and creative thinking.

M3:To provide integrated technology and management theories to augment worldwide business practices.

M4:To foster for Industry-Institute Interaction for Research and Development in the emerging fields and creating opportunities for employability and entrepreneurship

M5:To promote lifelong learning through exposure to the corporate world and by upholding ethical values and Social Responsibility to enrich the lives of mankind.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs) :

MBA programme curriculum is designed to prepare the post graduate students

- I. To have a thorough understanding of the core aspects of the business.
- II. To provide the learners with the management tools to identify, analyze and create business opportunities as well as solve business problems.
- III. To prepare them to have a holistic approach towards management functions.
- IV. To inspire and make them practice ethical standards in business.

PROGRAMME OUTCOMES (POs):

PO1: Business Knowledge:

An understanding of management skills in their areas of specialization in fields of Finance, Human Resources, Marketing, Production, Supply Chain and Logistics, Healthcare, Banking and Insurance enabling enriched global business impact

PO2 Critical Thinking and Problem Analysis:

Ability to Identify, Formulate and Analyze the contemporary issues in the business process by critical thinking to identify the best strategic alternatives.

PO3 Conduct investigations of complex Problems:

Conduct investigation using appropriate research design and analyze the qualitative and quantitative data by suitable research tools and techniques to solve the complex problem with detailed interpretation.

PO4 Strategic Decision Making and solutions:

Apply technology and Brain Storm the innovative ideas to enhance organizational efficiency and choose the optimal strategic decision for obtaining effective solutions

PO5 Modern tool usage:

Ability to utilize the advanced contemporary tools and techniques for predictive analysis and modeling of the complex business issues to sustain and flourish in the dynamic business environment.

PO6 Entrepreneurial Skills

Ability to business ideas, develop business plans, understand regulatory requirements, and locate sources of finance to start, manage and sustain new business ventures

PO7 Environment and sustainability:

Ability to understand the business environment and develop legal and ethical strategic plan that align

with societal and environmental context for the purpose of sustainable development in the business.

PO8 Professional Integrity:

Ability to address the social, legal, cultural diversity issues through ethical and value based.

PO 9 Team work:

Ability to function effectively on multi-disciplinary teams with extensive participation

PO 10 Communication:

Ability to communicate effectively and clearly, to comprehend, write, prepare and make effective presentations.

PO11 Global Business Opportunities:

Ability to demonstrate the management principles and understand the diverse opportunities in the management functions and apply those as an Entrepreneur, Projects, field Work, Consultancy, and work in the Multidisciplinary business environment.

PO12 Life-long learning:

Ability to empathize the impact of changing management concepts and thoughts on society and create awareness of contemporary issues through independent and lifelong learning.

PROGRAMME SEPCIC OBJECTIVES (PSO):

PSO 1: To have a thorough understanding of the core aspects of the business.

PSO 2: To provide the learners with the management tools to identify, analyze and create business opportunities as well as solve business problems.

PSO 3: To prepare them to have a holistic approach towards management functions.



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BONAFIDE CERTIFICATE

This is a certified Bonafide Record Work of Mr./Ms. _____

Register No. _____ submitted for the Semester End Examination

University Practical Examination held on _____ in **AGEM 302 –**

SEARCH ENGINE OPTIMIZATION LABRATORY during the year 2026-2027.

Signature of the Lab In-charge

Head of the Department

Internal Examiner

External Examiner

DATE _____

AGEM302 – SEARCH ENGINE OPTIMIZATION

SEARCH ENGINE OPTIMIZATION Laboratory							
Program me & Branch	BE/B.TECH – CSE, IT, AIDS & ECE	Sem.	Category	L	T	P	Credit
Prerequisites		5	Minor	3	0	0	3
Practical Experiments 1. Keyword Research and Competitive Analysis Using SEO Tools 2. On-page Optimization for Blog posts and Web pages 3. Backlink Analysis and Link-Building Strategies 4. Technical SEO Audit and Implementation 5. Monitoring and Improving page Speed SEO Performance Total: 30 periods							
COURSE OUTCOMES: By the end of this course, learners will be able to:						Bloom's Taxonomy Level	
CO1	Master the principles and techniques of search engine optimization.					K1	
CO2	Effectively conduct keyword research and implement on-page optimization strategies.					K2	
CO3	Analyze and improve website performance through technical SEO practices.					K1	
CO4	Develop and execute link-building strategies to enhance Website authority.					K1	
CO5	Utilize SEO tools to monitor, measure, kind optimize search rankings					K2	

AGEM302 SEARCH ENGINE AND OPTIMIZATION LAB

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EX NO: 1	KEYWORD RESEARCH AND COMPETITIVE ANALYSIS USING SEO TOOLS
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Aim:

To perform keyword research and competitor analysis using SEO tools.

Objectives

- Identify relevant keywords.
- Analyze competitor websites.
- Compare keyword metrics.
- Prepare a simple content plan.

Tools Required

- Google Keyword Planner
- Ubersuggest
- SEMrush/Ahrefs
- Google Trends
- MS Excel or Google Sheets

Theory

Keyword research identifies words users search for online. Competitive analysis studies competitor websites to find their keywords, rankings, and SEO strategies. This helps select effective keywords and create better content for higher search engine rankings.

Example

Keyword	Search Volume	Difficulty
SEO Tips	15,000	Medium
Keyword Research	10,000	Low
SEO Tools	20,000	Medium

Example

Website: Online Book Store

Keyword	Search Volume	Difficulty
SEO Tips	15,000	Medium
Keyword Research	10,000	Low
SEO Tools	20,000	Medium

Procedure

1. Open an SEO tool.
2. Enter a seed keyword.
3. Collect keyword suggestions.
4. Check search volume and difficulty.
5. Analyze top competitors.
6. Select suitable keywords.
7. Prepare a content plan.

Sample Content Plan

Content Title	Primary Keyword
SEO Basics	SEO Tips
Keyword Research Guide	Keyword Research
Best SEO Tools	SEO Tools

Observation

- Long-tail keywords have lower competition.
- High-quality competitor content ranks better.
- Search intent affects keyword selection.

Result

Keyword research and competitor analysis were completed successfully. Suitable keywords were selected, and a basic content plan was prepared to improve SEO performance.

EX NO: 2	ON-PAGE OPTIMIZATION FOR BLOG POSTS AND WEB PAGES
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Aim:

To optimize a blog post or web page using on-page SEO techniques.

Objectives

- Learn on-page SEO basics.
- Optimize titles, headings, keywords, and images.
- Improve search engine ranking and readability.

Tools Required

- Google Chrome
- Google Keyword Planner
- MS Word/Google Docs
- Yoast SEO or Rank Math
- Google Search Console

Theory

On-page SEO is the process of optimizing webpage content and HTML elements to improve search engine rankings. It includes optimizing the title, meta description, URL, headings, keywords, images, and internal links to make content more visible and user-friendly.

Example**Topic:** Digital Marketing Benefits

- **Keyword:** Digital Marketing Benefits
- **Title:** 10 Benefits of Digital Marketing for Small Businesses
- **URL:** /digital-marketing-benefits
- **Meta Description:** Learn how digital marketing helps small businesses grow online.

Procedure

1. Select a topic.
2. Research keywords.
3. Create an SEO-friendly title and URL.
4. Write content using headings and keywords.
5. Add optimized images and links.
6. Check SEO score and publish.

Sample Content Plan

Item	Example
Topic	Digital Marketing Benefits
Primary Keyword	Digital Marketing Benefits
Title	10 Benefits of Digital Marketing
URL	/digital-marketing-benefits
MetaDescription	Benefits of digital marketing for businesses.

Observation

- Keywords were used naturally.
- Title, headings, and URL were optimized.
- Images included ALT text.
- Readability and SEO score improved.

Result

The webpage was successfully optimized using on-page SEO techniques, improving search engine visibility, content quality, and user experience.

EX NO: 3	BACKLINK ANALYSIS AND LINK-BUILDING STRATEGIES
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Aim:

To analyze website backlinks and implement effective link-building strategies to improve SEO performance.

Objective

- Analyze the backlink profile of a website.
- Identify quality and toxic backlinks.
- Compare competitor backlinks.
- Develop ethical link-building strategies.

Tools Required

- Google Search Console
- Ahrefs
- SEMrush
- Moz Link Explorer
- Ubersuggest
- Microsoft Excel / Google Sheets

Theory

Backlinks are links from one website to another and are an important SEO ranking factor. High-quality backlinks improve domain authority, search rankings, and website traffic. Common link-building methods include guest blogging, content marketing, broken link building, and directory submissions.

Example: Website: <http://www.greenleaforganics.com>

Metric	Value
Backlinks	1,250
Referring Domains	180
Domain Authority	42
Toxic Links	20

Procedure

1. Open a backlink analysis tool (e.g., Ahrefs or Google Search Console).
2. Enter the website URL.
3. Analyze backlinks, referring domains, and toxic links.
4. Compare competitor backlink profiles.
5. Plan and implement suitable link-building strategies.
6. Monitor backlink growth regularly.

Sample Content Plan

Week	Content	Activity
1	Blog Post	Guest Posting
2	Infographic	Social Sharing
3	Case Study	Resource Page Links
4	How-to Guide	Broken Link Building

Observation

- High-quality backlinks improved website authority.
- Toxic backlinks reduced SEO performance.
- Competitor analysis revealed new backlink opportunities.

Result

Backlink analysis helped evaluate the website's SEO strength and identify improvement areas. Ethical link-building strategies increased domain authority, search visibility, and organic traffic.

EX NO: 4	TECHNICAL SEO AUDIT AND IMPLEMENTATION
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Aim:

To perform a technical SEO audit and improve website performance.

Objective

- Identify technical SEO issues.
- Improve crawling, indexing, and page speed.
- Enhance website performance.

Tools Required

- Google Search Console
- Google PageSpeed Insights
- Screaming Frog SEO Spider
- Google Lighthouse
- GTmetrix

Theory

Technical SEO improves a website's crawlability, indexing, speed, mobile usability, HTTPS, and overall search engine performance.

Example

Issue: Slow page speed and broken links.

Solution: Optimize images, fix broken links, and submit an XML sitemap.

Procedure

1. Analyze the website.
2. Check page speed and mobile usability.
3. Find SEO issues using audit tools.
4. Fix errors (links, metadata, sitemap).
5. Test and monitor improvements.

Sample Content Plan

Audit Item	Action
Page Speed	Optimize images
Broken Links	Fix URLs
XML Sitemap	Create & submit
Meta Tags	Add missing tags

Observation

Technical issues were identified and corrected, improving website performance.

Result

The website achieved better crawlability, faster loading, and improved SEO performance.

EX NO: 5	MONITORING AND IMPROVING PAGE SPEED FOR SEO PERFORMANCE
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Aim:

To monitor and improve website page speed for better SEO performance.

Objective

- Measure page speed.
- Identify performance issues.
- Apply speed optimization techniques.
- Improve SEO and user experience.

Tools Required

- Google PageSpeed Insights
- GTmetrix
- Google Lighthouse
- Chrome Browser

Theory

Page speed is the time a webpage takes to load. Faster websites improve user experience, reduce bounce rate, and achieve better search engine rankings.

Example

A webpage load time improved from 5.5 s to 2.2 s after image compression and code optimization.

Procedure

1. Test the website using PageSpeed Insights.
2. Identify speed issues.
3. Optimize images and minify CSS/JavaScript.
4. Enable browser caching.
5. Retest and compare the results.

Sample Content Plan

Item	Details
URL	www.example.com
Initial Speed	5.5 s
Optimization	Image compression, Minification, Caching
Final Speed	2.2 s

Observation

- Faster page loading.
- Higher performance score.
- Better user experience.

Result

The webpage speed was improved successfully, resulting in enhanced SEO performance and faster website loading.

EX NO: 6:	LOCAL BUSINESS KEYWORD RESEARCH
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Aim: Perform keyword research for a local business and identify suitable primary, secondary, and long-tail keywords.

Tools Required:

- Google Keyword Planner
- Ubersuggest
- Google Trends
- MS Excel / Google Sheets

Procedure:

1. Choose a local business category.
2. Enter relevant seed keywords in an SEO tool.
3. Record search volume, competition/difficulty, and keyword intent.
4. Identify at least five long-tail keywords.
5. Compare two competing websites and prepare a keyword list.

Observation:

- Long-tail keywords generally provide more specific search intent.
- Keyword selection should match the target audience and business objective.

Result:

Local business keywords were identified and compared successfully.

EX NO: 7	KEYWORD INTENT CLASSIFICATION
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Aim: Classify keywords according to informational, navigational, commercial, and transactional search intent.

Tools Required:

- Google Keyword Planner
- Google Trends
- Ubersuggest
- MS Excel

Procedure:

1. Prepare a list of 20 keywords for a selected topic.
2. Examine the wording and likely user purpose.
3. Classify each keyword by search intent.
4. Compare search volume across intent categories.
5. Select keywords appropriate for different stages of the customer journey.

Observation:

- Search intent is important when selecting keywords.
- Different intent categories require different content formats.

Result: Keywords were successfully classified according to search intent.

EX NO: 8

COMPETITOR KEYWORD GAP ANALYSIS

Aim: Perform a keyword gap analysis between a selected website and two competitors.

Tools Required:

- SEMrush/Ahrefs
- Google Search
- MS Excel

Procedure:

1. Select one target website and two competitors.
2. Collect ranking keywords for each website.
3. Identify keywords used by competitors but not by the target website.
4. Filter keywords by relevance and difficulty.
5. Prepare a priority list for future content.

Observation:

- Competitor keyword gaps reveal content opportunities.
- Relevant keywords should be prioritized over keywords with high volume alone.

Result:

A competitor keyword gap analysis was completed and content opportunities were identified.

EX NO: 9

KEYWORD MAPPING AND CONTENT PLANNING

Aim: Map selected keywords to website pages and prepare an SEO-oriented content plan.

Tools Required:

- Google Keyword Planner
- Google Trends
- MS Excel / Google Sheets

Procedure:

1. Collect relevant keywords.
2. Group similar keywords by topic.
3. Assign primary and secondary keywords to suitable pages.
4. Define search intent for each group.
5. Prepare a content calendar with page title and target keyword.

Observation:

- Keyword grouping reduces content duplication.
- Each important page should have a clear primary keyword and search intent.

Result:

Keywords were mapped to suitable pages and a basic content plan was prepared.

EX NO: 10	SEO OPTIMIZATION OF A PRODUCT PAGE
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Aim: Optimize a product/service page using title, headings, keywords, URL, meta description, and image ALT text.

Tools Required:

- Google Chrome
- Google Keyword Planner
- MS Word/Google Docs
- Yoast SEO or Rank Math

Procedure:

1. Select a product or service.
2. Research the primary keyword and related terms.
3. Create an SEO-friendly title and URL.
4. Write a meta description and structured headings.
5. Add suitable ALT text and review readability.

Observation:

- Relevant keywords should be used naturally.
- Clear headings and descriptive ALT text improve usability and search visibility.

Result:

The product/service page was successfully optimized using on-page SEO techniques.

EX NO: 11

INTERNAL LINKING OPTIMIZATION

Aim: Improve the internal link structure of a website to support navigation and SEO.

Tools Required:

- Google Chrome
- Google Search Console
- Screaming Frog SEO Spider
- MS Excel

Procedure:

1. Select a website or sample website.
2. Identify important pages and existing internal links.
3. Find pages with few or no internal links.
4. Add relevant contextual links between related pages.
5. Check anchor text and verify that links work correctly.

Observation:

- Relevant internal links help users discover related content.
- Descriptive anchor text provides useful context.

Result:

Internal links were identified and improved successfully.

EX NO: 12**IMAGE SEO OPTIMIZATION**

Aim: Optimize website images for SEO, accessibility, and page performance.

Tools Required:

- Google Chrome
- Google PageSpeed Insights
- Image compression tool
- MS Word/Google Docs

Procedure:

1. Select images for a sample webpage.
2. Use descriptive file names.
3. Resize and compress suitable images.
4. Add suitable ALT text and captions where appropriate.
5. Check page performance before and after optimization.

Observation:

- Compressed images can reduce page load time.
- ALT text should describe the meaningful content or purpose of an image.

Result:

Images were optimized for SEO, accessibility, and page performance.

EX NO: 13**BLOG POST SEO AUDIT**

Aim: Audit an existing blog post and improve its SEO and readability.

Tools Required:

- Google Chrome
- Google Keyword Planner
- Yoast SEO or Rank Math
- Google Docs

Procedure:

1. Select an existing blog post.
2. Identify its primary keyword and search intent.
3. Review title, headings, keyword usage, URL, links, and meta description.
4. Improve readability and add relevant internal/external links.
5. Compare the page before and after optimization.

Observation:

- SEO and readability should be considered together.
- Content should satisfy search intent rather than simply repeat keywords.

Result:

The blog post was audited and improved successfully.

EX NO: 14

BACKLINK PROFILE COMPARISON

Aim: Compare the backlink profiles of a target website and two competitor websites.

Tools Required:

- Ahrefs
- SEMrush
- Moz Link Explorer
- Ubersuggest
- MS Excel

Procedure:

1. Enter the target website and competitors into a backlink tool.
2. Record backlinks and referring domains.
3. Compare authority indicators where available.
4. Identify strong referring domains.
5. Prepare a comparison table and identify opportunities.

Observation:

- A larger number of backlinks does not always mean better quality.
- Relevant and authoritative referring domains are valuable.

Result:

The backlink profiles were compared and major opportunities were identified.

EX NO: 15

BROKEN LINK BUILDING EXERCISE

Aim: Identify broken links on relevant websites and prepare an ethical broken-link-building plan.

Tools Required:

- Ahrefs/SEMrush
- Google Search
- MS Excel

Procedure:

1. Choose a relevant topic.
2. Find resource pages with broken links.
3. Verify the broken URL.
4. Create or identify useful replacement content.
5. Prepare a professional outreach plan for the website owner.

Observation:

- Broken link building should provide genuine value.
- Replacement content must be relevant to the original resource.

Result:

Broken-link opportunities were identified and an ethical outreach plan was prepared.

EX NO: 16

GUEST POSTING OPPORTUNITY ANALYSIS

Aim: Identify suitable websites for ethical guest-posting opportunities.

Tools Required:

- Google Search
- Ahrefs/SEMrush
- Moz Link Explorer
- MS Excel

Procedure:

1. Define a target topic.
2. Search for relevant industry websites accepting contributions.
3. Evaluate relevance and authority indicators.
4. Record contact/page details and topic opportunities.
5. Prepare a shortlist of suitable websites.

Observation:

- Relevant websites are more valuable than unrelated high-authority sites.
- Guest content should be original, useful, and relevant to the audience.

Result:

Potential guest-posting opportunities were evaluated and shortlisted.

EX NO: 17**LINK BUILDING CAMPAIGN PLAN**

Aim: Prepare a four-week ethical link-building campaign for a selected website.

Tools Required:

- Ahrefs/SEMrush
- Google Search
- MS Excel/Google Sheets

Procedure:

1. Select a website and target topic.
2. Define linkable content assets.
3. Choose suitable outreach activities.
4. Prepare a four-week schedule.
5. Define metrics for monitoring campaign progress.

Observation:

- Consistent, relevant outreach supports sustainable link acquisition.
- Campaign results should be monitored using measurable indicators.

Result:

A structured four-week link-building campaign plan was prepared.

EX NO: 18**XML SITEMAP AUDIT AND SUBMISSION**

Aim: Audit an XML sitemap and identify improvements required for search engine crawling.

Tools Required:

- Google Search Console
- Google Chrome
- XML sitemap file

Procedure:

1. Locate the XML sitemap.
2. Check whether important URLs are included.
3. Identify invalid, redirected, or duplicate URLs.
4. Correct sitemap issues.
5. Submit or resubmit the sitemap through Google Search Console where applicable.

Observation:

- A sitemap should contain useful, canonical URLs.
- Regular sitemap monitoring can help identify technical issues.

Result:

The XML sitemap was audited and improvements were identified.

EX NO: 19

MOBILE FRIENDLINESS AND CORE WEB VITALS CHECK

Aim: Evaluate a webpage for mobile usability and performance-related SEO issues.

Tools Required:

- Google PageSpeed Insights
- Google Lighthouse
- Google Chrome

Procedure:

1. Select a webpage.
2. Run a mobile performance test.
3. Record key performance and usability findings.
4. Identify issues affecting loading or mobile experience.
5. Prepare recommendations and retest.

Observation:

- Mobile usability is important for user experience.
- Performance improvements should be verified through retesting.

Result:

The webpage was evaluated for mobile usability and performance issues.

EX NO: 20**CANONICAL URL AUDIT**

Aim: Check canonical URL implementation and identify duplicate-content risks.

Tools Required:

- Google Chrome
- Google Search Console
- Screaming Frog SEO Spider

Procedure:

1. Select a website.
2. Crawl or inspect important pages.
3. Record canonical URLs.
4. Identify missing, conflicting, or incorrect canonical references.
5. Prepare corrective recommendations.

Observation:

- Canonicalization helps indicate the preferred version of similar pages.
- Canonical references should be consistent with the intended indexable URL.

Result:

Canonical URL issues were identified and corrective recommendations were prepared.

Aim: Audit robots.txt directives and evaluate whether important pages can be crawled.

Tools Required:

- Google Chrome
- Google Search Console
- Screaming Frog SEO Spider

Procedure:

1. Locate the robots.txt file.
2. Review allowed and disallowed paths.
3. Identify whether important pages are unintentionally blocked.
4. Check crawl behaviour using audit tools.
5. Prepare recommended corrections.

Observation:

- Robots.txt controls crawler access but does not replace other indexing controls.
- Important pages should not be unintentionally blocked.

Result:

The robots.txt configuration was audited and crawl-control recommendations were prepared.

EX NO: 22	IMAGE COMPRESSION AND PAGE SPEED IMPROVEMENT
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Aim: Measure page speed before and after image compression and compare the results.

Tools Required:

- Google PageSpeed Insights
- GTmetrix
- Google Lighthouse
- Chrome Browser

Procedure:

1. Select a webpage containing images.
2. Record the initial performance result.
3. Compress and resize suitable images.
4. Retest the webpage.
5. Compare initial and final results.

Observation:

- Large images can contribute to slow page loading.
- Optimization should balance image quality and performance.

Result:

Image optimization was completed and page-speed results were compared.

EX NO: 23

CSS AND JAVASCRIPT OPTIMIZATION

Aim: Identify opportunities to reduce the impact of CSS and JavaScript on webpage performance.

Tools Required:

- Google PageSpeed Insights
- Google Lighthouse
- Chrome DevTools

Procedure:

1. Run a performance audit.
2. Identify render-blocking or oversized resources.
3. Review CSS and JavaScript loading.
4. Apply suitable optimization recommendations.
5. Retest and compare performance results.

Observation:

- Efficient resource loading can improve page performance.
- Changes should be tested to ensure functionality is maintained.

Result:

CSS and JavaScript performance opportunities were identified and evaluated.

EX NO: 24

BROWSER CACHING AND PERFORMANCE AUDIT

Aim: Evaluate browser caching and related performance opportunities for a webpage.

Tools Required:

- Google PageSpeed Insights
- Google Lighthouse
- Chrome DevTools

Procedure:

1. Select a webpage.
2. Run a performance audit.
3. Review caching-related recommendations and network resources.
4. Identify suitable caching improvements.
5. Retest the webpage and record the results.

Observation:

- Caching can reduce repeated resource downloads.
- Performance should be measured before and after changes.

Result:

Caching-related performance issues were evaluated and improvement measures were documented.

EX NO: 25**BEFORE AND AFTER SEO PERFORMANCE REPORT**

Aim: Prepare a performance report showing the effect of technical optimization on page speed.

Tools Required:

- Google PageSpeed Insights
- GTmetrix
- Google Lighthouse
- MS Excel

Procedure:

1. Select a sample webpage.
2. Record initial performance metrics.
3. Apply at least two suitable optimizations.
4. Retest using the same tool and conditions.
5. Prepare a before-and-after table and summarize the improvement.

Observation:

- Using the same measurement method improves comparability.
- Performance reports should clearly document the optimization performed.

Result:

A before-and-after page-speed performance report was prepared successfully.